DataScout

"Don't search for datasets. Describe what you need."

AI-Powered Public Dataset Discovery & Verification

Team No. 2

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The Dataset Search Problem

Finding the right public dataset can be difficult and time-consuming for data science and machine learning projects.

01 Manual Discovery

Users search across multiple public dataset portals and sources

02 Requirement Mismatch

A dataset may appear relevant but fail key requirements: domain, location, features, target, size, or constraints.

03 Verification Gap

Finding a dataset is not enough. Users need to verify whether the actual dataset satisfies their requirements.

04 Decision Uncertainty

Users must manually compare candidate datasets and determine which one is actually suitable.

> **"Finding a dataset is not enough. You need evidence that it fits."**

From Dataset Search to Dataset Intelligence

DataScout is an AI-powered cloud platform designed to simplify and automate public dataset discovery and verification for data science and machine learning projects.



"AI interprets. Deterministic logic verifies."

Ranked Recommendations

Suitability score and requirement-wise evidence for every candidate dataset.

PASS / PARTIAL / FAIL Evaluation

Transparent grounded explanations with clear per-requirement verdicts.

Usage & Next-Step Guidance

Dataset access guidance, starter code, preprocessing recommendations and suggested next steps.

How DataScout Works

01 Describe

User enters their project idea and requirements in natural language.
e.g. "I need an Indian dataset for crop yield prediction in Maharashtra with rainfall and temperature data."

02 Understand

AI Requirement Analyzer extracts: domain, geographic region, required features, target variable, dataset size and constraints.

03 Validate

Requirement Schema Validation checks that extracted requirements are properly structured before dataset retrieval.

04 Discover

DataScout searches supported public sources including data.gov.in, Hugging Face and additional public APIs.

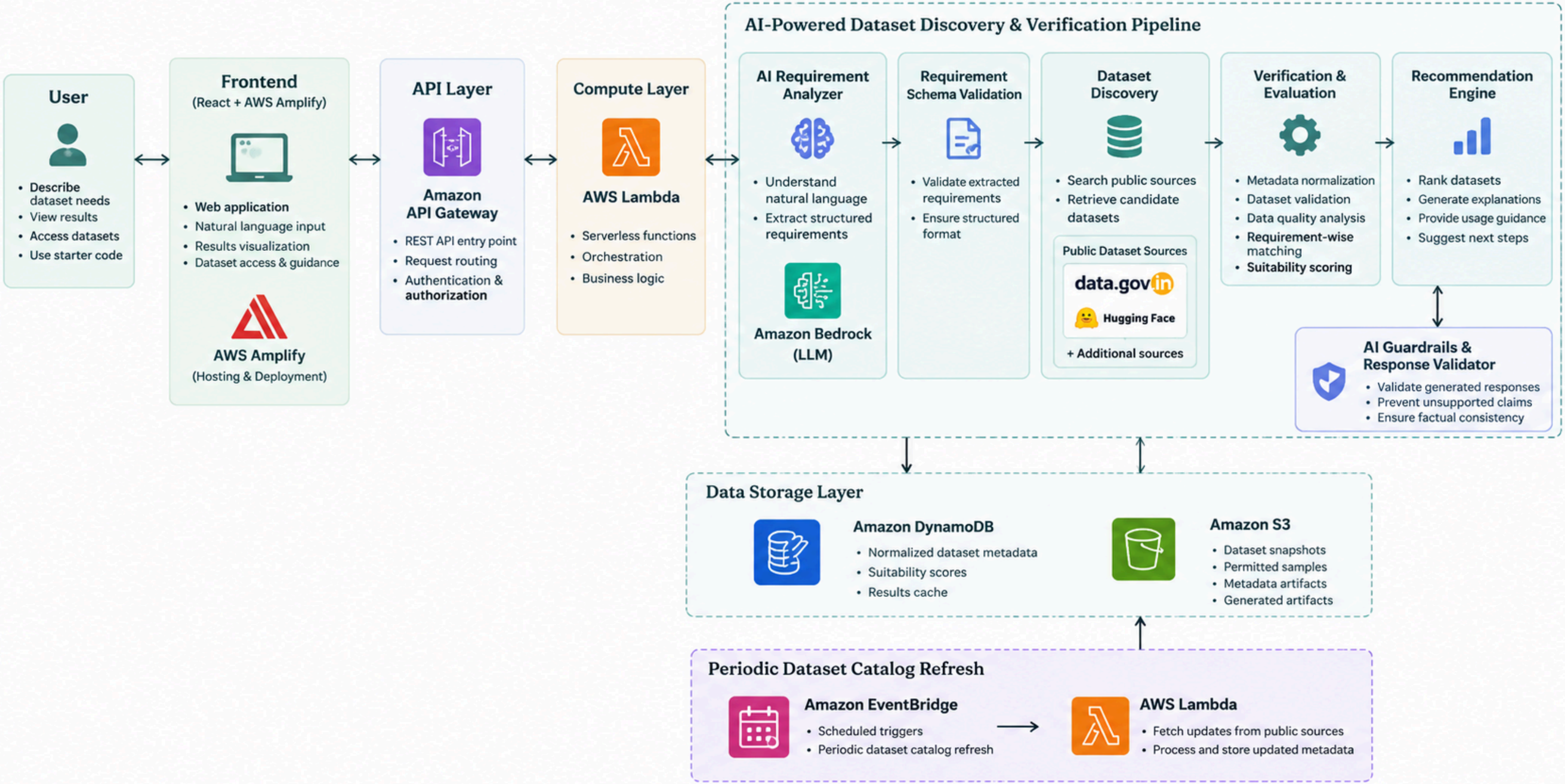
05 Verify

Retrieved datasets go through metadata normalization, dataset validation and data quality analysis.

06 Recommend

DataScout calculates suitability, ranks candidates, provides evidence, generates grounded explanations and usage guidance.

DataScout — System Architecture



AI That Explains, Logic That Verifies

AI Layer

Natural-language requirement interpretation
Requirement extraction
Grounded explanations
Usage guidance

Verification Layer

Schema validation
Dataset validation
Quality analysis
Deterministic suitability scoring
Requirement-wise evidence

"LLM ≠ Source of Truth"

Requirement Interpretation

The LLM is used for natural-language requirement interpretation and generating grounded explanations — not for determining dataset suitability.

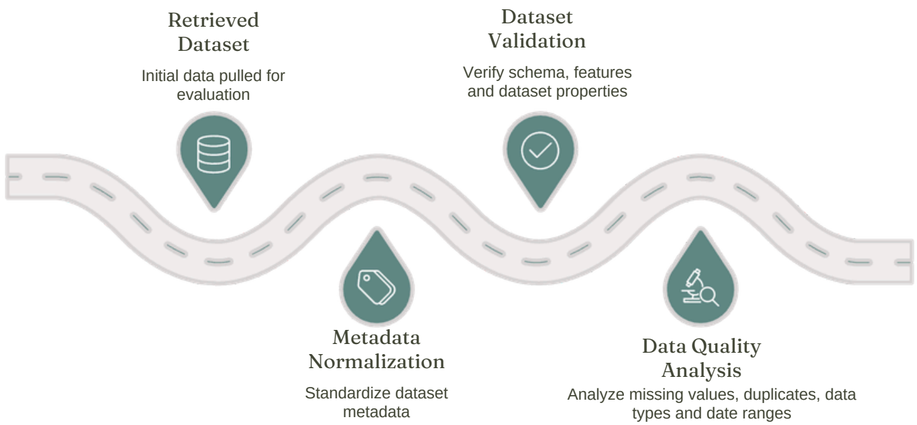
Deterministic Verification

Actual dataset verification and suitability scoring are performed using retrieved dataset information and deterministic processing.

AI Guardrails & Response Validation

Before results reach the user, the AI Guardrails & Response Validator checks the generated response so unsupported claims and hallucinated dataset information are not presented as facts.

Verification Before Recommendation



Requirement 1

PASS : Requirement met based on verified dataset metadata.

Requirement 2

PARTIAL : Requirement partially satisfied; evidence provided.

Requirement 3

FAIL : Requirement not met; transparent explanation included.

The Suitability Scorer performs deterministic matching between the user's validated requirements and verified dataset information, producing a reproducible suitability score with requirement-wise evidence.

> **"Recommendation is based on evidence, not just relevance."**

AWS Services Used



AWS Amplify

Frontend hosting and deployment for the React web application.



Amazon API Gateway

REST API entry point between the frontend and the backend processing layer.



AWS Lambda

Serverless backend processing and orchestration across all pipeline components.



Amazon Bedrock

Powers AI capabilities including natural-language requirement interpretation and grounded explanations.



Amazon DynamoDB

Stores normalized dataset metadata, suitability scores and results cache information.



Amazon S3

Stores dataset snapshots, permitted samples, metadata artifacts and generated artifacts.



Amazon EventBridge

Triggers periodic dataset catalog refresh to keep discovery results current.

Why DataScout?

Literature Review

Traditional Dataset Portals

Strength: Large collections with search/filter mechanisms
Gap: Users must manually verify requirements

Dataset / Semantic Search

Strength: Metadata and semantic relevance beyond keywords
Gap: Relevance ≠ requirement satisfaction

Dataset Recommendation Systems

Strength: Rank datasets by relevance and metadata
Gap: Ranking lacks requirement-wise verification evidence

LLM / Agent-Based Assistance

Strength: Natural-language interaction and flexible interpretation
Gap: Recommendations can lack grounding in verified data

DataScout Differentiation

Workflow comparison

Existing	**DataScout**
Search → Retrieve → Rank	Describe → Extract → Validate → Discover → Verify → Score → Explain

Key differentiators

Requirement-Wise Verification

Each requirement evaluated as PASS/PARTIAL/FAIL with evidence

Deterministic Suitability Scoring

Verified data + deterministic matching, not LLM-driven

Multi-Source Normalization

Common metadata structure for consistent comparison

Dataset Validation & Quality Analysis

Schema, features, coverage, and quality indicators evaluated

Grounded Recommendations

AI explanations from verified results with guardrails

DataScout addresses the gap between dataset retrieval and dataset verification by combining natural-language requirement understanding with deterministic, requirement-wise verification and suitability scoring.

Challenges & Limitations

Technical Challenges

Data Quality Inconsistency: Public datasets vary in completeness, accuracy, and cleanliness.

Schema Variability: Sources use different structures, formats, and field naming conventions.

API Rate Limits: External source restrictions can affect discovery speed and coverage.

Catalog Freshness: Dataset metadata and availability may change between scheduled refreshes.

Project Limitations

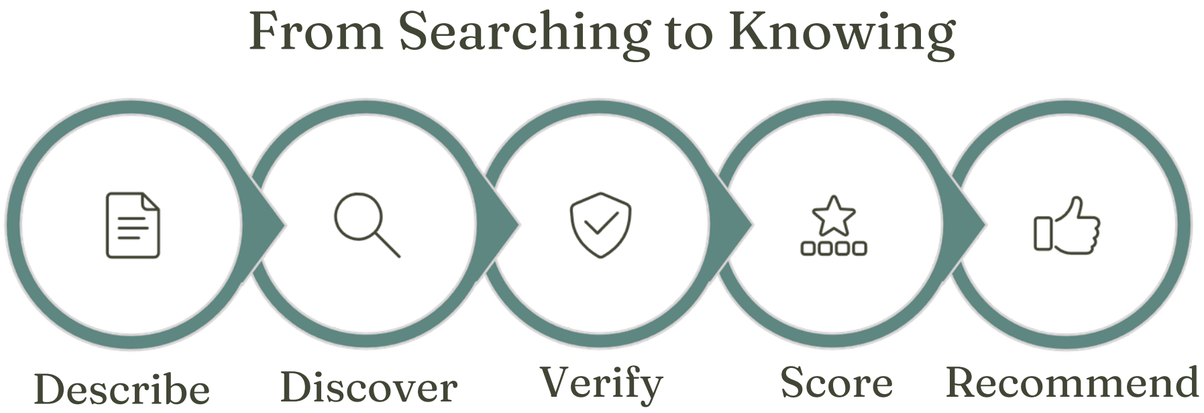
Public Sources Only: Current scope is limited to publicly accessible datasets.

Verification Depends on Accessibility: Verification depth depends on available metadata and data samples.

LLM Interpretation Dependency: Requirement extraction depends on model capabilities; verification and scoring remain deterministic.

New-Source Integration Cost: Supporting new or obscure sources requires additional connector and normalization effort.

Conclusion :



DataScout combines Generative AI, deterministic data verification, public dataset sources and cloud infrastructure to transform dataset selection into a transparent, evidence-based and requirement-driven workflow.

References:

[1] Chapman et al., Dataset Search: A Survey, VLDB Journal, 2020.
[2] Paton et al., Dataset Discovery and Exploration: A Survey, ACM Computing Surveys, 2023.
[3] Viswanathan et al., DataFinder: Scientific Dataset Recommendation from Natural Language Descriptions, ACL 2023.
[4] Chen et al., Leveraging Large Language Models for Dataset Discovery, ASIS&T, 2025.